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Problem Chosen
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2023 HiMCM
Summary Sheet

Electrifying a Metro Bus Fleet Without Reducing Service

Question and baseline. We study conversion of TransLink's fixed-route bus service to fully electric operation by 2033 subject to preserving recovered fall-2023 departures and capacity proxies. Existing trolleybuses count as electric; ordinary renewal is the comparator.

Model. We specify trip, capacity, winter energy and charger constraints; the city inventory remains a proportional scenario without a joint dispatch certificate. Gross discounted cost is separated from payments after 0%, 25% or 50% grants; emissions use a shared annual service ramp and explicit partial boundaries. A global Pareto frontier requires plant data that are absent, so we report a named decomposition and do not label infeasible points as an optimum.

Observed result. The recovered fall-2023 GTFS contains 54,497 trips in 25 service patterns. The representative weekday has 1,752 source-declared blocks and 1,087 simultaneous trips; annual fixed-route service is 5.58 million hours and 106.87 million km. A fuel- and stock-weighted tailpipe proxy gives 126–408 tonnes NO_x per year. Vehicle identity, age, depot and site power remain unresolved.[2, 13, 42, 43, 44]

Conditional inventory and cost scenario. At a 300-km overnight-charge sensitivity, the generic non-deck case has 1,612 net new battery positions. The integrated class envelope purchases 1,647 route BEBs plus 262 one-for-one trolley renewals (1,909 electric vehicles), ends with 1,927 electric and zero conventional fixed-route assets, and costs CAD 4.71 billion versus CAD 2.69 billion for renewal plus operations (-75.0%). The high class case is CAD 5.26 billion (-95.5%), and the 0/25/50% grant scores are $-75.0/-42.8/-10.5\%$. Its partial-production, operation and maintenance boundary yields 4.6–42.7% emissions reduction. A 700-kWh double-decker leaves 17 of 57 duties short of one overnight charge; a 150-kWh shuttle leaves 49 of 101. London's 17-year renewal peak is 776 vehicles per year.[9, 17, 13, 23, 25]

Recommendation. Start depot power and grid upgrades in 2024, preserve the recovered service patterns, and procure proven 40-/60-ft blocks first. Gate double-deck and shuttle conversion on in-route charging, all-electric winter tests and verified site capacity.

Keywords: electric bus; fleet replacement; service preservation; charging; life-cycle cost.

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1 Problem definition and study boundary

The competition asks for environmental and financial consequences, a ten-year transition in three metropolitan areas, and a decision-maker letter.[1] We take 31 December 2023 as Vancouver's decision baseline and plan purchases and construction in 2024–2033. Two evidence layers are reported separately: observed 2023 counts define the service floor, while a retrospective 2025 fleet snapshot and later public project envelopes reconstruct missing class/age context and bound future costs. No realized later purchase is presented as a 2023 decision. Metro Vancouver is the primary case; its census metropolitan area had 2,642,825 residents in 2021, exceeding the problem's 500,000-person threshold.[5] Chicago and London test transferability using their separately dated 2023 fleet snapshots; London's March audit is not treated as December stock.

Prior fleet models optimize electric vehicle count and mix jointly with charging and scheduling,[34] use a multi-period purchase, salvage and demand-charge formulation,[35] or solve multi-depot time-window and recharge constraints.[36] This study connects those ideas to a recovered public timetable, a capacity proxy where the vehicle roster is absent, a normal-renewal comparator, and a separately stated emissions boundary. The result is also a data-validity test: a missing garage or winter record changes the status to `UNVERIFIED` rather than becoming a zero-cost assumption.

The problem's definition requires propulsion *and accessories* to use a zero-emission electric source. Thus a battery bus with a diesel auxiliary heater is not fully electric in this study. TransLink's historical depot-BEB cost table assumes precisely such a heater at 0.02 L/km; its 1.4 kWh/km electricity figure cannot be carried over unaltered to an all-electric winter bus. Renewable diesel, renewable natural gas and hybrids do not satisfy the endpoint. Hydrogen fuel cells are treated as a separate zero-tailpipe option because eligibility would require a certified zero-emission hydrogen supply. Already electric trolleybuses satisfy the endpoint, provided their wired operation remains feasible. TransLink has a 2040 zero-emission target, making 2033 an accelerated counterfactual.[9, 17]

The decision unit is year y, bus class k, depot d, trip i, vehicle v and charger interval t. Classes include standard, articulated, highway coach, double-deck, community shuttle and trolley vehicles. We preserve 2023 fixed-route coverage and do not offset electrification cost by reducing service.

2 Evidence and baseline audit

The 2025 budget's 2023 actual service table includes SeaBus in its published transit subtotal. We subtract 13,713 SeaBus hours, 185,370 km and 71,367,323 capacity km to obtain the fixed-route bus values in Table 1; the source's five operating-group rows reconcile exactly in the accompanying audit. The implied service-km-weighted average capacity is $5{,}715{,}791{,}401/106{,}868{,}324 = 53.48$ passenger spaces per bus-km, an aggregate calibration target rather than a standard-bus specification. The earlier statutory report rounds bus hours to 5,577,000, yielding a slightly different 98.013% route coverage check.[13, 29]

The fall GTFS is preserved at commit `eb584b1` of an independent public mirror, and every local source file matches its immutable Git blob. Its first-stop distance field is absent for every trip, so exact trip length is unknown; a monotonic shape-distance envelope gives a median interval width of 0.432 km, about 4.3%, and a maximum 26.309 km. We use the upper bound only for conservative energy screening and never fill the exact field. The annual CSV has 191 route identifiers while

2023 observation	Value	Source/status
Published route identifiers	191	official CSV[19]
Fall GTFS bus trips; service patterns	54,497; 25	`23SEP_20231020`[2]
Maximum service day trips; source blocks	22,492; 1,756	8 September 2023[2]
Representative weekday blocks; simultaneous trips	1,752; 1,087	14 November 2023[2]
Route service hours	5,466,190.5	official CSV[19]
Actual fixed-route bus hours, excluding SeaBus	5,575,772	2023 actual[13]
Actual fixed-route bus service km	106,868,324	same source[13]
Actual fixed-route bus capacity km	5,715,791,401	same source[13]
Route-file share of exact bus hours	98.035%	calculated
Route/time-bin frequency rows; nonmissing	6,261; 6,041	official CSV[30]
Route/direction/time-bin load rows; nonmissing	11,659; 9,371	official CSV[31]
Revenue-fleet operating GHG, all modes	112,591 t CO_2e	2023 report[32]
CNG fleet count	299	2023 report[32]
Trolley plus battery-electric buses	280	Dec 2023 notice[7]
Battery-electric buses	about 5	Dec 2023[8]
Trolley buses	262	Feb 2023 notice[6]
2022 detailed CMBC fleet snapshot	1,711	historical anchor[33]
Derived 2023 fleet total	1,725 (1,711–1,733)	bridge estimate[33]
2025 active CMBC fleet	1,733; 22 battery, 261 trolley, 1,450 other	18 Nov 2025[33]

Table 1: The recovered fall timetable supplies source-declared trips and blocks. The annual route table and fall feed use different version and aggregation scopes.

the feed has 232 route short names with trips because the annual table groups some branches and omits discontinued service. Both are retained with that version difference. The GTFS does not supply vehicle ids, ages, depot links, battery state or an as-operated record. Route and arrival data are used by permission of TransLink, which assumes no responsibility for their accuracy or currency.[2]

The 2018 official service guidelines define peak load factor as average peak passengers divided by practical vehicle capacity. Their reference capacities are 24 (mini-bus), 47 (standard trolley), 50 (standard bus or coach), 70 (articulated trolley), and 75 (articulated bus). Dividing each positive 2023 time-bin load by its load factor reproduces one of these values in 8,705 of 9,371 cells; a further 344 cells give exactly 85, consistent with TransLink's separately described 80–86-seat double-deck vehicles. Thus 9,049 cells (96.56%) match six capacity clusters, and 322 do not. Across 180 routes with positive cells, 34 show more than one cluster or an unmatched cell. This is a useful *aggregate vehicle-class diagnostic*, not a trip-level vehicle assignment: capacities are historical averages, the 85-person class is inferred, and mixed or differently averaged cells cannot establish an individual bus's capacity. The reproducible cell-level calculation and every unmatched value are retained in the data supplement.[20, 21, 31]

Capacity must also be checked on the replacement vehicle. Current TransLink operating limits are 20 passengers for community shuttles, 77 for 40-ft buses and 120 for articulated buses; the 2018 planning values are lower and distinguish seating from practical capacity.[26, 20] Manufacturer examples span 68 passengers for a Nova LFSe+, 88 for a current 40-ft Xcelsior CHARGE NG, 118 for the 2023 XE60 specification and 123 for the current XE60. A smaller electric bus therefore requires another departure vehicle or a different model when the baseline departure has 77 or 120 spaces.[27, 28]

The 2023 accountability report also records 112,591 tonnes CO_2e of *revenue-fleet operating* emissions, comprising 110,402 Scope-1 and 2,189 Scope-2 tonnes; it separately reports 33,718 tonnes of biogenic CO_2 from renewable fuels. The revenue-fleet number covers bus and non-bus modes, excludes vehicle and station manufacture, and reflects that year's renewable-fuel mix (96% of purchased bus natural gas was renewable). It is therefore an independent accounting check and an upper bound on the included bus operating inventory, not a substitute for the 2024–2033 same-service, life-cycle normal-renewal baseline.[32]

A later 18 November 2025 CMBC allocation table provides a retrospective fleet-profile check: 1,733 active buses at seven sites, including 261 trolleys, 22 battery buses, 535 hybrids, 365 CNG, 379 diesels and 171 gasoline shuttles; 63 more are in storage. Its manufacture-year counts reconcile to 1,733 and imply 392 buses aged 0–4 years, 543 aged 5–9, 91 aged 10–14, 705 aged 15–19 and two older than 20. It is not substituted for the 2023 register. Combined with the observed 2023 trolley and zero-emission counts, we select 1,725 for a stock scenario and 1,711–1,733 for sensitivity; these are not observed stock or proven bounds. About 285 vehicles were built in 2023–2025; a strict 2033 endpoint would retire many of them at age 8–10, so the cost model must include early replacement.[33, 7]

The 2020 TransLink fleet transition plan records 1,672 CMBC vehicles, including 188 standard and 74 articulated trolleybuses.[9] These are historical values, not a verified 2023 fleet register. The February 2023 notice gives 262 trolleys, while a December official article says approximately 275 trolleys and five BEBs. A same-day release gives only their combined count, 280. Subtracting February's 262 from December's 280 would imply 18 BEBs, but this crosses reporting dates and conflicts with the explicit December article. The retained count of 18 is only a scenario parameter,

not evidence of deliveries.[6, 7, 8] A June 2023 Board report lists 2023–24 procurement projects for 46 articulated hybrids, 15 standard BEBs, 50 CNG buses, 168 community shuttles and 25 double-deck buses; it records the 25 double-deck buses as accepted on a Q2 2023 timeline, while the other group totals were mostly outstanding. These are procurement states, not a year-end fleet roster.[11] A valid 2033 purchasing table still requires an age, class and depot register.

A separate June 2023 TransLink presentation places four BEBs and 262 trolleys in service *at that date*. It gives screening prices of approximately CAD 1.2 million per BEB and CAD 0.46–0.61 million per on-route charger. Its then-current capital concepts put Hamilton Route 100 charging at CAD 2.3 million, Port Coquitlam plus ten on-route chargers at CAD 46 million, and a Marpole depot sized for 340 BEBs at CAD 500 million. A newly built 500–600-bus-equivalent depot was screened at CAD 1.2–1.8 billion including land and development. These are 2023 planning envelopes with different scopes, *not* delivered item prices or additive line items; their commissioning dates changed in later plans. They supply bounded infrastructure cases while site-specific electrical capacity and construction cash flow remain missing.[12, 17]

A 2025 capital-program audit narrows the age uncertainty: 188 standard trolleys entered service in 2006–07 and 73 articulated trolleys plus one pilot were expected to retire in 2028–29. It identifies projects tied to 57 diesel-hybrids and 50 CNG buses reaching end of life in 2023, 126 diesel-hybrids in 2024, 36 diesel-hybrids and nine highway coaches in 2025, and 39 articulated diesels in 2026. The plan specifies 57 BEBs for 57 retiring hybrids, seven BEBs for six retiring hybrids, and 45 articulated BEBs for 39 retiring diesels in three separate programs. These are retrospective planned cohorts, not confirmed 2023 stock, deliveries, or a route-feasible 2033 solution. The 57-BEB project's CAD 106.846 million gross capital budget includes CAD 102.498 million of reported external funding; divided by 57, its *gross project budget* is CAD 1.874 million per planned bus. The other identified BEB projects yield CAD 1.867 million per planned standard bus and CAD 3.037 million per planned articulated bus on the same budget/count arithmetic. These are project envelopes, not delivered bus-only unit quotes. The accompanying cohort audit reproduces the ratios and divisions from the published table.[13]

As a bounded funding illustration using only the 57-BEB project's gross budget, assuming *all* CAD 106.846 million is grant-eligible, 0%, 25% and 50% hypothetical grants leave agency payments of CAD 106.846, 80.135 and 53.423 million, respectively. The gross project budget remains CAD 106.846 million in each case. These are undiscounted funding splits of one later-reported program, not incremental economic scores: eligibility, spending dates, and the same-service conventional renewal budget are unresolved. The actual CAD 102.498 million external-funding entry is a distinct retrospective budget fact and does not override the problem's 50% scenario limit.[13]

A subsequent 2026 plan changes the retrospective picture further. A named project now proposes 39 60-foot diesel buses, 64 in-route-charged 40-foot BEBs and 153 depot-charged 40-foot BEBs for 236 retiring diesels: 256 planned replacement vehicles, or 1.085 per retirement. Another cohort proposes 31 BEBs for 27 retiring hybrids, a 1.148 ratio. The 2025 plan's 45 articulated BEBs for 39 retiring diesels is an earlier description with possible cohort overlap; its count is *not* added to the later project. The 2025 plan separately names 50+84+30=164 CNG replacements for 2023–2025 retirement cohorts. Thus these distinct named programs plan $164 + 39 = 203$ non-electric replacement vehicles associated with 2023–2026 retirement cohorts. If procured and still operating, all 203 would need accelerated removal to achieve the strict 2033 endpoint. This is a planning-exposure count, not a verified delivery count or a complete 2033 fleet requirement.[13, 15]

3 Fleet and service model

3.1 Decision variables and normal-renewal baseline

Let x_{ykda} be buses purchased into class k, depot d and age cohort a; r_{ykda} old vehicles retired from that cohort; h_{ydq} charger positions of power type q; and z_{iv} equal one when vehicle v performs scheduled trip i. Let c_{vt} denote charge energy and e_{vt} battery energy. For ordinary renewal, age-expired conventional buses are replaced with conventional equivalents while the *same calendar service pattern and passenger-capacity demand* is served. Electric candidates are compared with that feasible renewal path. Historical purchase expenditure is sunk; early retirement enters through foregone residual or sale value.

For each class and depot, stock obeys

$$N^p_{y,k,d,a} = N^p_{y-1,k,d,a-1} + x^p_{y,k,d,a} - r^p_{y,k,d,a}, \quad p \in \{\text{electric}, \text{conventional}, \text{trolley}\}. \tag{1}$$

No diesel, hybrid, CNG or hydrogen propulsion or non-electric accessory may operate on any fixed-route trip in 2033. Retirement and purchase are not constrained one-for-one: needed vehicle counts emerge from schedules and spare ratios. A short-range vehicle may force additional buses, chargers or drivers.

3.2 Hard service and technical constraints

For every calendar service pattern p, every scheduled trip $i \in I_p$ is retained at its route, direction and departure time. Multiple smaller buses may serve one departure when necessary, and each assigned class must be compatible:

$$\sum_{v \in V_i} z_{iv} \geq 1 \qquad (i \in I), \tag{2}$$

$$\sum_{v \in V_i} Q_v z_{iv} \geq Q^0_i \qquad (i \in I), \tag{3}$$

$$\sum_{i \in I_{r,s,t}} \sum_{v \in V_i} Q_v z_{iv} \geq Q^0_{r,s,t}. \tag{4}$$

The final expression denotes capacity available at the original route–direction peak; the trip-level inequality protects each departure. A smaller bus can replace a larger one only if an additional vehicle on that departure restores the capacity and both vehicles pass scheduling, energy and site checks. Accessible trips require accessible vehicles; depot/class spare stocks are protected. If demand growth is modeled, it is applied equally to baseline and electric cases.

The source-declared block is the second hard unit. For consecutive trips $i \prec j$ in block b, retained on one vehicle, $s_j - e_i \geq \tau^{road}_{ij} + t^{turn}_{ij}$. If splitting is necessary, both vehicles must be able to reach a documented exchange or charging site without moving a departure. In the fall feed, 879 of 1,752 weekday blocks interline more than one route; none of the 20,679 consecutive transfers has zero gap across more than 0.5 km, supporting GTFS blocks as duty hypotheses. Most long blocks cannot recharge midday without replacement or terminal charging; distance divided by range is only a screen, not a vehicle count.

As a separate *calibration check*, the schedule covers 22,492 bus trips on the maximum service day, while the annual output should reproduce 5,575,772 fixed-route bus service hours, 106,868,324

service km and 5,715,791,401 capacity km within documented measurement differences. The capacity-km subtotal detects systematic undersupply, but cannot substitute for departure, block and site constraints.[2, 13]

The executed range screen uses

$$d_b^U = \sum_{i \in b} \ell_i^U, \qquad n_{bk}^{(j)} = \left\lceil \frac{d_b^U}{R_k^{(j)}} \right\rceil, \tag{5}$$

where $R_k^{(j)}$ is an assumed reliable range and $n_{bk}^{(j)}$ is the number of sequential duty requests. This is a conditional distance screen, not a vehicle-count bound. A route exchange, depot return or charger restriction can only increase the physical requirement, while sharing split duties between blocks may reduce it; the result is therefore not a purchase count.

Each vehicle's assigned trips and charging intervals cannot overlap. Location-to-location deadhead times must fit between events, including departure from and return to its depot. Driving to or from service also removes a vehicle from the available depot spare count. For battery vehicles, energy follows

$$e_{v,t+1} = e_{vt} + \eta_q c_{vt} - \epsilon_{vk}^{\text{traction,winter}} (\ell_{it} + \ell_{vt}^{\text{deadhead}}) - E_{vt}^{\text{accessories}}, \tag{6}$$

$$\underline{s}_v B_v \le e_{vt} \le \bar{s}_v B_v. \tag{7}$$

Here $E_{vt}^{\text{accessories}}$ includes all-electric cabin heating and other onboard equipment during trips, deadhead, charging and active layovers; the traction coefficient excludes that energy to prevent double counting. An unmeasured winter accessory load makes an electric schedule `UNVERIFIED`, not zero-energy. Battery capacity and passenger capacity are linked when battery weight reduces available seats or payload. Charger occupancy, power and opening year impose

$$\sum_v u_{vdt} \le H_{dt}, \quad \sum_v P_{vdt} \le P_{dt}^{\text{grid}}, \quad H_{dt} = 0 \text{ until commissioned.} \tag{8}$$

Conventional baseline vehicles instead require a fuel budget, with jointly calibrated distance and elapsed-duty rates a_v, b_v in a consistent fuel unit (L or kg):

$$F_{v,t+1} = F_{vt} - a_v \ell_{vt} - b_v \Delta t, \qquad F_v^{\text{reserve}} \le F_{vt} \le F_v^{\text{tank}}. \tag{9}$$

The implemented check assumes no refuelling within a duty; the time rate includes stationary consumption and must not duplicate the distance rate. Missing fuel data leave the normal-renewal comparator unverified. Such vehicles may serve intermediate years but cannot satisfy the strict 2033 endpoint.

Trolleybuses use a separate wired-network check. Each trip, deadhead and powered layover needs a certified connected path and an operating feeder. Concurrent demand must obey $\sum_{v \in V_{ft}} P_v \le P_f^{\text{firm}}$, where firm power is available after other loads and reserve. The code conservatively reserves each vehicle's measured all-in peak draw on every feeder traversed for the entire moving segment; stationary accessories use their separate demand. A failure of this sufficient power certificate calls for finer feeder timing before declaring physical infeasibility. Missing path or feeder records leave existing trolley operation unverified, without assigning these vehicles fictitious batteries or buying their capacity again.

Depot bus parking capacity, charger footprint, charging efficiency, construction lead time and route-specific winter testing must also pass. A charger site may be the home depot or a route terminal, provided the location and travel-time checks pass. The accompanying validator checks the supplied trip, energy and site records, returning `UNVERIFIED` for missing essential inputs; it cannot certify a city-wide solution from time-bin averages.

3.3 Explicit assumptions and validation

Five assumptions define the evidence boundary. (A1) Every recovered 2023 fall departure and direction is a service obligation; the representative weekday scales fleet demand. (A2) The values 24, 47, 50, 70, 75 and 85 are capacity proxies; 47 and 70 become trolley-route proxies only through the independent 262-trolley count. (A3) Reliable range is a technology scenario, and the screen from 149 to 350 km is not a delivered-battery guarantee. (A4) The unverified 1,725-bus scenario and 18 assumed existing BEBs seed the central stock table, with 5–18 BEBs retained as a sensitivity. (A5) Public vendor and project envelopes span different nominal years; their 3% result is price-mixed screening PV. Validation uses source-total reconciliation, immutable GTFS hashes, interval distances, an independent charging LP bound, zero-gap London MILPs, a 432,768-case named-cell reproduction, and 2033 stock identities. All scenario and endpoint values are regenerated by the local pipeline.

3.4 Optimization and executed decomposition

The full problem is

$$\min C^{\text{gross}} \quad \text{s.t.} \begin{array}{l} \text{service, block and capacity;} \\ \text{SOC, charger and feeder;} \\ \text{spare, inventory and endpoint.} \end{array} \tag{10}$$

followed by the ϵ-constraint family $\min C^{\text{gross}}$ subject to $G \le \epsilon$ for the Pareto frontier. The station and depot register is insufficient for that global integer program. We therefore execute a documented decomposition: (i) audit every calendar service pattern and block; (ii) screen alternate reliable ranges; (iii) price a fixed-service accelerated replacement scenario; and (iv) compute matched production, operation and maintenance emissions. The validator classifies absent inputs as `UNVERIFIED`. This separated approach follows the joint fleet/scheduling logic of Rogge et al., the multi-year transition logic of Pelletier et al., and the multi-depot timing logic of Gkiotsalitis et al.[34, 35, 36]

4 Costs, funding and environmental accounting

Let K_y include bus and battery procurement (initial battery included in a quoted vehicle price), chargers, depot civil work, grid connection and distribution upgrades, construction/design, training and any extra buses. Let O_y include electricity, fuel, maintenance, replacement batteries, drivers, demand charges and regular charging-site upkeep. Let S_y be realized salvage and R_{2033} the remaining value of vehicles and durable infrastructure at the horizon. A complete appraisal requires one price basis and a matching real or nominal discount rate. The executed result below uses a stated 3% discount as scenario arithmetic on public envelopes spanning 2021–2026 nominal CAD; it is therefore called price-mixed screening PV, not a constant-dollar bid appraisal.

$$C^{\text{gross}} = \sum_{y=2024}^{2033} \frac{K_y + O_y - S_y}{(1+r)^{y-2023}} - \frac{R_{2033}}{(1+r)^{10}}, \tag{11}$$

$$\Delta C = C^{\text{gross}}_{\text{electric}} - C^{\text{gross}}_{\text{renewal}}. \tag{12}$$

For grant rate $f \in \{0, 0.25, 0.50\}$ and eligible expenditures E_y, $C^{\text{agency}} = C^{\text{gross}} - \sum_y fE_y/(1+r)^{y-2023}$. A grant changes who pays, not gross buses, copper, electricity or labour. Local employment and procurement impacts require an additional regional economic model and are reported separately to avoid double counting.[37]

The regional layer records direct local procurement L_y, wages W_y and job-years J_y. Any induced value added uses a regional input–output multiplier m_y on labour and supplier value added, $V_y^{\text{local}} = m_y(L_y + W_y)$, not on the entire gross construction outlay. Feasible charging capacity, reliability and terminal asset value enter the fleet model through served trips, avoided extra vehicles, bill savings and residual value. Construction spending is not counted again as a benefit.

Define a life-cycle greenhouse-gas inventory over equal service:

$$G = G_{\text{vehicle+battery}} + G_{\text{infrastructure}} \tag{13}$$

$$+ \sum_y \left(G_{\text{fuel,y}} + G_{\text{electricity,y}} + G_{\text{maintenance,y}}\right) - G_{\text{end of life}}. \tag{14}$$

Electricity uses an explicitly selected grid-intensity scenario. British Columbia's 2023 integrated-grid factor is 0.0113 kg CO_2e/kWh; it is an annual average and should be varied against marginal-generation possibilities.[38] On-route NO_x, particles, electricity use and battery-production impacts appear in separate columns rather than being concealed by a single carbon score. If compatible manufacturing/infrastructure factors are absent, the *life-cycle* score is unavailable even when operating emissions can be estimated.

The requested scores are

$$S_C = 100\frac{C_{\text{renewal}} - C_{\text{electric}}}{C_{\text{renewal}}}, \qquad S_G = 100\frac{G_{\text{renewal}} - G_{\text{electric}}}{G_{\text{renewal}}}. \tag{15}$$

Negative values remain negative. The economic score is based on *gross* cost, with subsidized agency cost shown beside it; otherwise the 50% grant would artificially change the societal merit ranking.

5 Observed and conditional results

5.1 Fall 2023 block and range screen

The recovered feed contains 54,497 bus trips, 25 calendar service patterns and 1,684,516 bus stop-time records after excluding rail, SeaBus and HandyDART. On 14 November 2023 it schedules 22,431 trips and 1,752 source-declared blocks. At the busiest instant, 1,087 trips are simultaneous. The largest service day, 8 September, has 22,492 trips in 1,756 blocks. These are planned duties, not an as-operated or vehicle registry.[2]

For energy screening, we classify blocks by the 2023 route–time capacity signature: 24, 47, 50, 70, 75 and 85 passenger-space proxies. The labels identify capacity only. Route blocks with 47/70 signatures are retained as a trolley-route proxy because the separate 2023 fleet count has 262 trolleys, but the GTFS contains no propulsion field. The 57 weekday 85-person double-deck blocks are excluded from the overnight-depot arithmetic. A 300-km scenario gives 1,741 sequential battery-block requests for 1,502 participating blocks, 239 extra requests (15.9%). At the 149/150-km electric-heat band, the corresponding requirement is 2,444 requests (62.7% more). Saturday

and Sunday produce larger ratios because fewer blocks cover long weekend duties; the weekday total sets the selected fleet scale.

Proxy class	Baseline blocks	Additional sequential block requests by range 150	185	250	300	350 km
40-ft battery proxy	1244	788	577	333	202	96
60-ft battery proxy	258	154	119	66	37	25

Table 2: 2023-11-14 service with 1,752 published duty blocks and 1,087 simultaneous trips. Capacity signatures are route-level proxies, not observed bus assignments. The 40-ft battery proxy contains minimum/standard capacity classes and 216 unclassified blocks; 47/70-seat signature blocks are retained only as a trolley-route proxy, and 85-person double-deck blocks are unassessed. Counts are daily sequential block requests, not a garage fleet or one-for-one bus count.

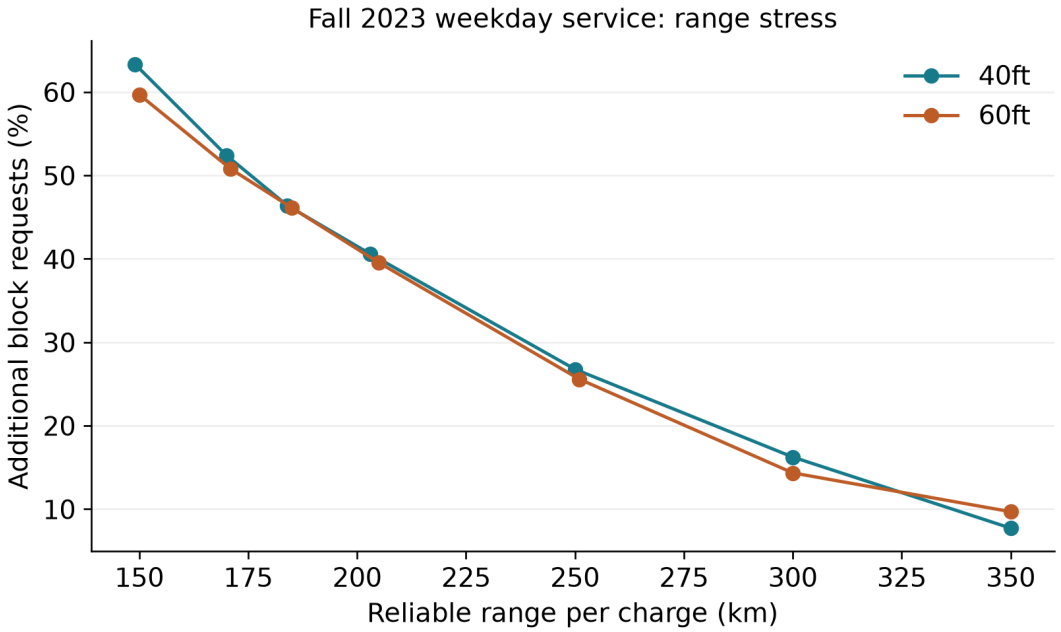


Figure 1: Distance divided by reliable range is a screening bound for extra sequential duty requests. It does not include route exchange time, garage return, charger occupancy or spare buses. Capacity signatures are not certified vehicle assignments.

Only 3 of the 20,679 weekday block transitions provide at least 60 minutes of gap; among the 4,850 route-changing transfers, only 2 do, and none provides 90 minutes. The blocks therefore mainly require overnight charging, terminal charging or a documented vehicle exchange. This supports retaining source blocks as duty hypotheses, while the missing garage and charger links prevent an operational pass.

5.2 Conditional ten-year cost and emissions

We evaluate an accelerated fixed-service policy case rather than relabel a full replacement as ordinary renewal. A uniform 17-year fleet renews 10/17 of its positions in a decade; the derived

1,388 non-deck fossil buses yield 816 baseline purchases. The 1,741 duty ratio is proportionally extrapolated onto an assumed 1,406-bus stock, selecting 1,630 positions. This is neither a lower nor upper vehicle bound: task overlap, location and reuse are unresolved. Retaining 18 assumed existing BEBs gives 1,612 net new generic positions. Vehicle prices use the 2023 CAD 1.2-million hardware screen in the lower case and the weighted 2025 project envelopes of CAD 1.874 million for a standard and CAD 3.037 million for an articulated bus in the upper case. The baseline uses the published 50-CNG-bus project envelope, CAD 0.923 million per planned vehicle.[12, 13, 14]

Depot infrastructure is priced at the 2023 Marpole screening ratio of CAD 500 million per 340 BEBs in the base case. The high case uses the 2024 CAD 6.52-billion screening envelope, which includes service expansion and multiple sites and is therefore a resource-cost sensitivity.[12, 17] Vehicles and infrastructure are spread over the implementation years; vehicle residual value follows a 17-year straight-line life. O&M applies only to the fossil-service share and ramps toward the BEB rate. Driver hours, already-electric trolley renewal and the 57 double-deck vehicles are outside this scored sub-boundary.

The larger purchase count already charges much of the early-retirement burden: 1,612 new batteries enter a 1,406-bus non-deck fleet before 2033, while the steady-state renewal path buys 816. Salvage and redeployment value of displaced young buses are not credited, so the base result is conservative on that item. Conversely, the high infrastructure envelope includes expansion not required by fixed 2023 service, so it is an upper-side sensitivity.

Scenario	Gross PV (CAD bn)	Score (%)
Conventional renewal	2.329	0
Electric, base infrastructure	4.575	-96.4
Electric, screening high	8.646	-271.2
Agency PV and score, grants 0/25/50% (base plan)		
	4.575 / 3.643 / 2.710	-96.4 / -56.4 / -16.4

Table 3: Conditional generic non-deck case. The 1,741 sequential duty requests are scaled to a derived 1,406-bus non-double-deck, non-trolley fleet under a 300-km range, giving 1,630 battery positions and 1,612 net purchases after 18 assumed existing BEBs. This is price-mixed screening arithmetic, not a constant-dollar bid appraisal. The 57 double-deck buses and 262 trolleys are outside this sub-boundary; the class-specific result adds them separately. The baseline renews 10/17 of unknown-age fossil positions, and vehicle residual follows a 17-year life.

The generic non-deck price-mixed present value is CAD 4.58 billion against CAD 2.33 billion for steady-state renewal, giving an economic score of -96.4%. The high generic infrastructure and vehicle case reaches CAD 8.65 billion and -271.2%. A 50% external grant changes the base agency payment to CAD 2.71 billion and the score to -16.4%; it improves affordability without reducing the project's resource cost. These figures are not a certified city optimum: the winter range is a 2020 sensitivity, depot power is absent, and the 2023 age roster is unavailable.

A class-specific inventory scenario selects 1,419 heavy non-deck positions, 57 double-deck buses and 189 community shuttles, with heavy-bus depot capacity sized from the 1,419 positions and class-specific vehicles and chargers added. The conversion buys 1,401 heavy, 57 deck and 189 shuttle battery replacements (1,647 in total). The ordinary baseline keeps the known 2019–23 double-deck cohort because none reaches a 17-year chassis life before 2033; early replacement is charged on the

Baseline operating mix	g CO_2e/km	Renewal (kt)	Electric (kt)	Score (%)
renewable fuel mix	302	376.2	373.5	0.7
rng heavy	960	960.4	636.4	33.7
weighted mix	1233	1202.8	745.4	38.0
fossil diesel	1580	1510.9	884.1	41.5

Table 4: Conditional generic non-deck ten-year emissions over the named boundary: vehicle and battery production, operating energy, and maintenance. Electric service rises from 10 to 100 percent over 2024–2033; remaining kilometres retain conventional operation and maintenance. Depot/charging construction and end-of-life credits are excluded. The renewable mix assumes 80% renewable-diesel reduction plus 96% RNG for CNG; RNG-heavy uses RNG but fossil diesel; weighted mix uses the 2020 propulsion mix. This is a separate boundary from the class-specific result.

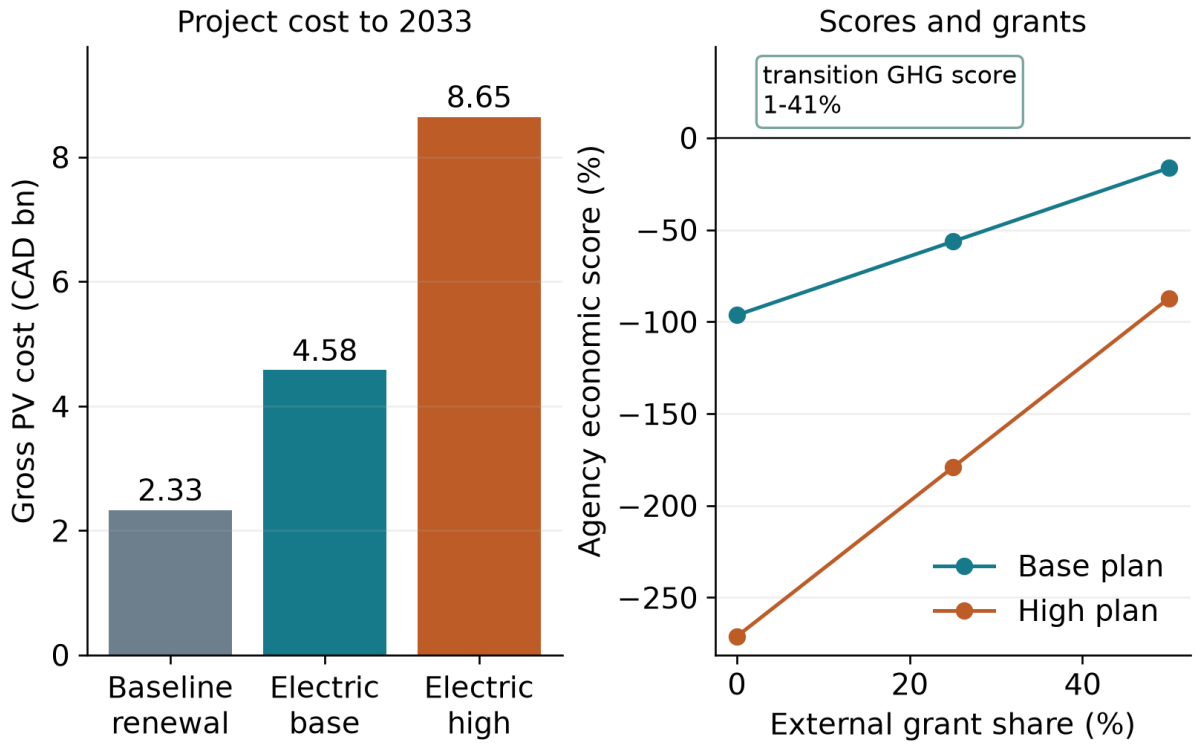


Figure 2: Conditional generic non-deck economics and funding; curves show agency scores. Gross cost remains negative-scored under grants; the resource cost does not disappear. The boxed 1–41% range is the generic non-deck production/operation/maintenance score, not the class-specific 4.6–42.7% envelope; both exclude depot construction and end-of-life credits.

electric side without a resale credit. Both paths renew 188 standard and 74 articulated trolleys under the public trolley replacement envelope (CAD 601 million nominal), so 262 common electric purchases preserve the trolley stock and do not change the comparison stock. Central price-mixed gross PV is CAD 4.71 billion versus CAD 2.69 billion for renewal and operations (-75.0%); 25/50% grants give $-42.8/-10.5\%$, while the high class envelope is CAD 5.26 billion (-95.5%). A 700-kWh double-decker leaves 17 of 57 weekday duties short of a single overnight charge; even an 827-kWh outlook leaves eight. A 150-kWh shuttle leaves 49 of 101 duties short. The 2033 endpoint therefore requires documented terminal or en-route charging and exceeds TransLink's later plan to begin double-deck and shuttle conversion after 2030.[11, 22, 23, 24, 25, 17, 13, 15]

A shared annual service ledger drives O&M and emissions: electric share $\alpha_y = (y-2023)/10$ leaves $1-\alpha_y$ of scored kilometres conventional. Use emissions are $D\sum_y[(1-\alpha_y)(g+0.0524)+\alpha_y(0.019+0.0675)]/1000$ tonnes. The generic case scores 0.7–41.5%, with 376–1511 kt in the baseline and 373–884 kt in the transition. For the class case, unknown shuttle bodies are omitted on both sides, removing 8.373 kt of inappropriate standard-bus body production from the baseline. Its 4.6–42.7% score corresponds to 17.7–680.1 kt avoided against 387–1592 kt baseline emissions. Both percentages cover scored conversion service, excluding common trolley and retained-BEB activity; they are not whole-fleet life-cycle scores. Standard-bus body proxies remain for heavy/deck vehicles; depot construction and end-of-life are excluded. The annual tailpipe proxy remains 126–408 t NO_x, 1.11–8.34 t $PM_{2.5}$ and 2.34–14.18 t PM_{10}; its fuel-by-km allocation is uncertain.[18, 17, 42, 43, 44]

At 1.37–2.0 onboard kWh/km and 90% charging efficiency, converted non-deck service needs approximately 135–197 GWh/year from the grid; the 1.6 kWh/km central value gives 158 GWh/year. The 1,612 new 335-kWh batteries contain about 31.3 kilotonnes of modeled LFP-production CO_2e using the transferred factor. These are scale checks, not a site load profile or a Vancouver battery specification.

As a historical benchmark, the route table still covers 98.035% of independently reported actual bus hours. The 2019-CAD and 2024-O&M reference cases retain their original narrower boundaries. These checks and the fixed-service policy case support the fleet decision; only an operator trip, garage, charger and age register can certify every 2033 duty.

For a transparent historical parameter check, the 2020 plan lists a 2019-CAD diesel bus at CAD 630,000, depot BEB at CAD 1,110,000, 50-kW charger position at CAD 298,000, 0.60 L/km diesel, 1.4 kWh/km electricity and 0.02 L/km heater fuel.[9] At 68,000 km/year the historical BEB uses 95,200 kWh plus 1,360 L of fuel, so its heater fails the strict 2033 endpoint. The CAD 298,000 charger quotation already includes design, permitting, project management, financing and contingency and must not be repriced. This is not a 2023 bid; TransLink's 1.15 depot replacement ratio also shows why one-for-one purchase is unsafe.

For a separate electric-only 40-foot reference, 2019-CAD lifetime costs are CAD 4.86/km for in-route BEB versus 4.62 diesel, 4.77 hybrid and 4.15 CNG; the three economic scores are -5.19%, -1.89% and -17.11%. These are single-vehicle comparisons, not fleet scores. Using 1.6 kWh/km, 0.60 L/km diesel, 2.630 kg CO_2e/L and the BC grid factor gives a 98.85% operational-only reduction, or 106.07 t CO_2e/year at 68,000 km. Vehicle identities, as-operated execution and site capacity remain unavailable, so this excludes manufacturing and fuel supply.[9, 10, 39, 38]

The 2024 ZEFTP offers a separate retrospective check. Its operating-and-maintenance costs are 3.30, 1.97, 2.77 and 2.28 CAD/vehicle-km for diesel, hybrid, CNG and BEB, respectively. On that *O&M-only* boundary, BEB scores are $+30.91\%$, -15.74% and $+17.69\%$ against the three alternatives; vehicle, battery and charger capital are absent. Its electric-heater BEB is reported

at 19 g CO_2e/km, versus 1580 for fossil diesel, 1420 for fossil hybrid, 1350 for fossil CNG and 250 for renewable CNG. Fuel mix therefore changes the operational environmental comparator. These 2024 values are neither 2023-known procurement inputs nor full life-cycle 2024–2033 totals; they cannot be added to the older 2019-CAD costs.[17]

An independent 2024 ICCT e-bus life-cycle method supplies transferable *sensitivity factors*: 6.6 kg CO_2e per kg of vehicle glider/powertrain, 58 kg CO_2e per kWh of LFP battery, and 52.4/67.5 g CO_2e/km for diesel/BEB maintenance. Its 12–15 m example uses a 335-kWh battery and glider masses from Latin American buses, so these are not observed TransLink specifications. Its own system boundary excludes charging infrastructure and bus recycling. We retain separate terms for these missing components and do not substitute this partial method for a completed Vancouver life-cycle inventory.[18]

Facility timing is also consequential. A December 2020 program listed 677 planned BEB purchases and CAD 246.4 million of depot/charging investments, but subsequent 2024 review targeted Port Coquitlam for over 100 BEBs in 2027 and Marpole for roughly 340 in 2028. It estimated six to eight years to deliver charging at existing transit centres and recommended commissioning six to twelve months before bus delivery. These are planning and later-review quantities, not 2023 installed charger counts. The 2024 screening capital envelope includes service expansion and has a stated -50% to $+100\%$ accuracy range, so assigning it wholesale to our fixed-2023-service scenario would misstate incremental cost.[10, 17]

The 2020 program arithmetic checks 677 planned BEB awards: CAD 236.9 million vehicle premium plus 246.4 infrastructure minus 96.3 operating savings gives CAD 387.0 million undiscounted. A later plan instead lists 84 CNG buses for a 2024 hybrid-retirement cohort, showing that the pathway was revised. These source-period amounts cannot be converted into 2024–2033 cost, grant or GHG scores without aligned dates and fleet stock.[10, 13]

The historical depot analysis supplies a direct *winter feasibility stress test*. With 660-kWh articulated batteries and fuel heaters, it projected depot-dependent replacement ratios of 1.13–1.55 and warned that electric-only heat could require roughly 10% more buses. Applied mechanically to the later capital project's 39 retiring articulated diesels, $\lceil 39(1.10)r_d \rceil$ gives 49–67 BEBs across hypothetical depot allocations, compared with its 45 planned depot-charged BEBs. The 4–22 difference is a *historical-technology sensitivity*, not a measured shortage: the 39 actual depot assignments, 2026 energy specifications and block schedules are unknown. It identifies a concrete winter trial and procurement question before certifying the strict 2033 endpoint.[9, 13]

5.3 Charging tariffs and the value of firm power

The original 2023 budget assumes diesel at CAD 1.66/L; this is a planning price, not an actual invoice. Its 32.17-million-L operational volume includes a broader system boundary and is not treated as fixed-route bus consumption.[3] BC Hydro's April-2023 tariff supplies energy, monthly demand and daily basic charges. RS1640 bills peak demand only during 06:00–22:00; RS1650's zero demand fee ends in March 2026 under that historical schedule. We therefore distinguish billed peak from physical grid power and do not extrapolate the temporary concession through 2033.[4]

For daily grid energy E, available charging hours T, site limit P, and rate-specific billed hours T_j, solve

$$\min M_j : \quad \sum_{t \in T} p_t \Delta t = E, \quad 0 \le p_t \le P, \quad p_t \le M_j \ (t \in T_j). \tag{16}$$

Power is zero outside T. If h_0 available hours are outside the billing window and $h_1 > 0$ are inside, the sharp lower bound is $M_j \geq \max\{0, (E - h_0 P)/h_1\}$. Physical feasibility also requires $E \leq (h_0 + h_1)P$; a wholly nonbilled feasible window gives $M_j = 0$. The executed CPU LP matches this independent bound in every feasible case. A second stage minimizes physical peak at the optimal billed peak. Monthly bills include energy, basic and demand charges, the prior-winter eleven-period minimum-bill floor, and the 1% rider refund; taxes and capital are excluded.

We normalize to 100 *distance-equivalent units*, each at the historical 68,000 km/year reference. These are not an observed fleet or a one-for-one replacement count. Historical onboard energy of 1.37–2.0 kWh/km, assumed charging efficiencies of 85/90/95%, three common return windows and 2.5/3.5/5-MW site limits give 243 scenarios. Annual energy is spread evenly across 365 days; this relaxation does not certify winter energy, battery capacity, charger-port occupancy or actual bus availability. A meter-measured intensity would not be divided by efficiency again.[9]

Rate	Physical MW	Billed MW	Annual CAD million
1600	4.140	4.140	1.350
1640	4.140	0.000	0.905
1650 (pre-Apr 2026)	4.140	4.140	1.113

Table 5: Conditional charging bounds for 100 distance-equivalent units, 68,000 km/year each, 1.6 battery-kWh/km, 90% charging efficiency, 22:00–06:00 availability and 5-MW site power. April-2023 prices are frozen for a 365-day comparison; bills include the 1% rider refund and exclude taxes and capital.

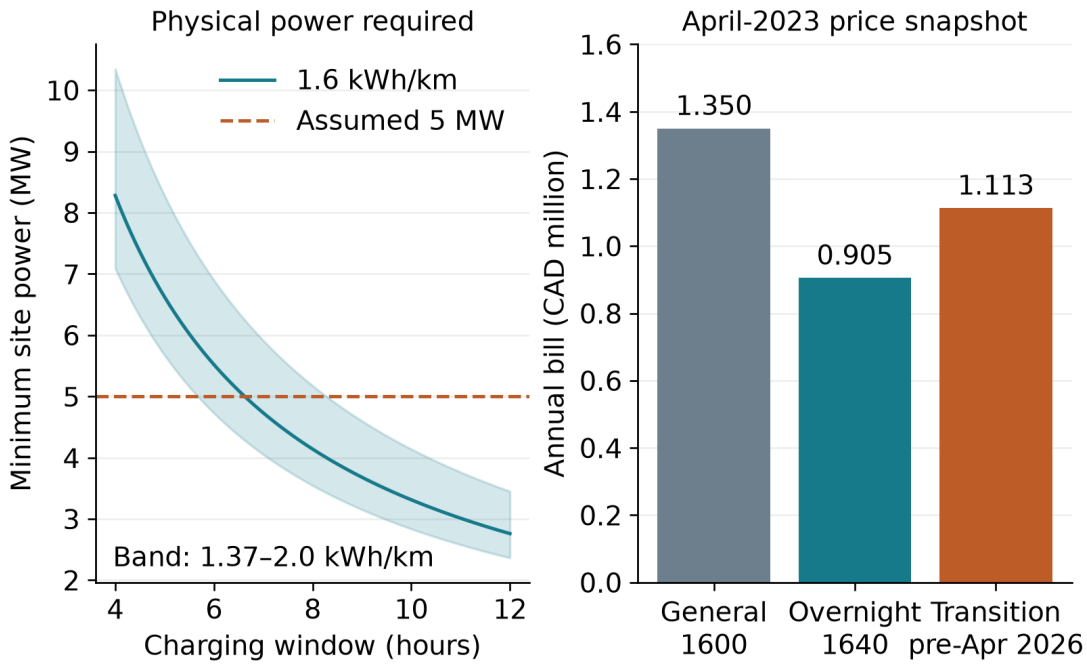


Figure 3: Physical charging-power bounds and conditional tariff bills. The left panel uses 90% charging efficiency; the right uses the central eight-hour case. A zero billed peak still requires 4.140 MW of physical power.

In the central case, an eight-hour window needs at least 4.140 MW, and RS1640 saves 32.96%

of the RS1600 electricity bill. At 3.5 MW, eight-hour-only charging is impossible under the declared energy requirement. Extending availability to 20:00–06:00 restores aggregate feasibility, but RS1640 then costs CAD 1.285 million/year versus CAD 1.227 million under RS1600. The bills cross at 3.598 MW; two separate LP solutions confirm this threshold. Between 3.312 and 4.140 MW, each extra firm kW saves CAD 594/year under RS1640 in this fixed-price, fixed-window case, before paying for the upgrade. This quantifies an infrastructure contribution through feasible service and avoided bills, without crediting construction expenditure as a second benefit. Individual vehicle certification and dated future tariffs remain necessary before using these bounds in a purchasing decision.

Depot	Active	Non-trolley	50-kW scale (MW)
VTC	430	169	8.45
HTC	289	289	14.45
BTCN	107	107	5.35
BTCS	161	161	8.05
STC	266	266	13.30
PTC	235	235	11.75
RTC	245	245	12.25
Total	1733	1472	73.60

Table 6: 2025-11-18 CMBC active-fleet scale by site. This is retrospective allocation, not 2023 stock; the power column is active non-trolleys times the historical 50-kW planning allowance, not available transformer capacity.

At the 2020 planning allowance of 50 kW per depot-charged bus, the generic 1,612 positions imply about 80.6 MW and the 1,647-position class conversion about 82.4 MW of aggregate connected capacity. TransLink's 2024 long-range estimate is 78 MW for full-fleet electrification by 2040. The 2023 accelerated plan therefore has roughly the same grid-scale challenge as the published 2040 path. This is a portfolio scale check, not a site transformer rating; only a feeder and interconnection study can determine 2033 feasibility.[9, 17]

6 Ten-year roadmaps for three metropolitan areas

The common transfer model is a stock-flow planning model: each year it buys compatible replacements, retires conventional stock, and reports the remaining electric and conventional inventories. Vancouver's richer public cost, energy and charger evidence then supplies the full class-specific financial and environmental application. Chicago and London use the same endpoint and purchase logic, while their unmatched tariff, contract and route records prevent a defensible transfer of Vancouver's dollar and emission factors. This is a deliberate second model, not a claim that a purchase count alone proves route feasibility.

The Vancouver column is an annual inventory path, not a garage roster. The integrated case buys 1,647 route BEBs plus 262 common trolley renewals across 2024–2033 (1,909 electric purchases, about 191 per year), retires the 1,445 conventional buses and ends with 1,927 electric assets and zero conventional rolling stock. The 262 trolley replacements are one-for-one renewals and leave the 262-position electric trolley stock unchanged. The 202 extra positions are proportional scenario

Year	Van. EV buy	Van. EV	Van. fossil	CTA buy	CTA fossil	TfL buy	TfL fossil
2024	204	446	1300	185	1665	775	6918
2025	203	611	1155	185	1480	776	6142
2026	204	776	1010	185	1295	776	5366
2027	202	941	865	185	1110	776	4590
2028	239	1106	720	185	925	776	3814
2029	201	1271	575	185	740	776	3038
2030	165	1436	430	185	555	776	2263
2031	164	1600	286	185	370	776	1493
2032	164	1764	142	185	185	776	724
2033	163	1927	0	185	0	776	0
Total	1909	1927	0	1850	0	7759	0

Table 7: Annual inventory path under three explicit planning models. Vancouver's 1,909 electric purchases comprise 1,647 route BEB positions plus 262 one-for-one trolley renewals; the stock endpoint remains 1,927 because replaced trolleys leave the fleet. Its 202 extra positions come from longer duty splits. Chicago purchases 1,850 electric replacements from a 1,875-bus inventory after 25 inferred electric buses. TfL renews 7,759 body-compatible positions under the 17-year scenario, including 7,693 non-electric conversions. Each model reaches zero non-electric stock in 2033; local garage and route feasibility is still required.

inputs, not proven consequences of duty splitting; 18 assumed BEBs seed the stock. The generic non-deck sensitivity instead uses 1,612 positions and excludes common trolley renewal. Because the published facility delivery range is six to eight years, depot design, utility applications and permits must begin in 2024 even if bus deliveries are phased later. A funded annual order near the tabulated level and a physically commissioned site must precede each service conversion.[17, 13, 15]

Chicago's year-end inventory is 1,875 buses, of which 25 are inferred from the June 2023 base order plus two earlier plug-ins. The inventory path buys 185 electric replacements per year, moving electric stock from 25 to 1,875 and conventional stock from 1,850 to zero by 2033. That schedule does not escape age risk: 1,056 buses are at least 13 years old and 539 at least 16. Under a 17-year chassis-life scenario, at least 539 positions are already near a 2024 renewal deadline; a 14-year scenario creates 1,056. Its matched fuel table yields annual diagnostics of 0.825 L diesel/km and 1.733 charging kWh/km, but covers only 85.55% of profile revenue miles. Garages, exact duties, local bids and a reconciled energy scope are required before the 185/year position becomes a procurement order.[45, 46, 47, 48]

London starts from 8,643 contracted positions, including 950 electric and 20 hydrogen. A 17-year cohort scenario renews 7,759 positions in total, converting all 7,693 non-electric (including hydrogen) slots, with a peak of 776 per year; the 14- and 12-year scenarios require 977 and 2,568 in their busiest years. The 12-year case is a timely-renewal stress scenario. London has the strongest public age register of the three, but its audit does not supply route blocks, depot power, winter energy or local vehicle prices, so its financial and environmental application remains unavailable.[49]

The later 2025 Investment Plan reports 4.424 million bus service hours in 2033, below the independently reported 2023 actual total. Because those accounting boundaries differ, we retain the

2023 service schedule as the hard floor rather than credit the transition with reduced operation; reconciling the later plan is a separate reporting task.[16, 13]

6.1 A solved age-cohort planning benchmark

For London, an original *fleet slot* denotes one available vehicle position of the same body type. Renewing a slot requires suitable electric capacity; the operational model may require additional vehicles. Let n_c be the observed size and b_c the registration year of cohort c, and choose a maximum chassis-life scenario L. These caps are assumptions for sensitivity, not asserted TfL rules. A non-electric cohort must renew by $d_c = \min(2033, \max(2024, b_c + L))$. An existing electric cohort enters the renewal model only when $b_c + L \leq 2033$; the 2013 Routemaster retains its chassis vintage despite its 2022 electric retrofit.[49]

Let integer a_{cy} allocate original slots to renewal years $2024 \leq y \leq d_c$. We first minimize the required peak annual renewal capacity M:

$$\min M : \qquad \sum_{y \leq d_c} a_{cy} = n_c, \quad \sum_{c} a_{cy} \leq M, \quad a_{cy} \in \mathbb{Z}_+. \tag{17}$$

With M fixed, a second solve minimizes $\sum_{c,y}(d_c - y)a_{cy}$, avoiding avoidable early renewal. Among ties, a third solve prioritizes earlier non-electric conversion without changing either optimum. This is an executed integer program using the public cohort counts, not an assumed uniform purchase schedule.

An independent certificate follows from $D_t = \sum_{c:d_c \leq t} n_c$:

$$M^\star = \max_{t=2024,\ldots,2033} \left\lceil \frac{D_t}{t - 2023} \right\rceil. \tag{18}$$

Necessity follows because all D_t slots must fit into $t - 2023$ annual capacities. Sufficiency follows by allocating the earliest-deadline cohorts first when every cohort may renew from 2024. Later technical release dates, class-specific production limits and charging construction can increase the requirement; the certificate concerns this cohort submodel.

Life cap	Peak slots/year	Existing EV renewals	Total slots
12	2,568	735	8,428
14	977	247	7,940
17	776	66	7,759

Table 8: Optimal London fixed-availability cohort benchmarks, 2024–2033. Life caps are scenarios. Counts are renewed original fleet slots; operational feasibility can require additional buses.

All three scenarios were solved on CPU using SciPy's HiGHS mixed-integer solver; every optimization phase reaches zero gap and the peak optima match the analytic bound. With a 17-year cap, 7,693 non-electric slots convert and 66 existing electric slots renew, requiring peak capacity 776 per year. The 14-year scenario requires 977 in its busiest year and renews 247 existing electric slots. The 12-year cap creates a 2,568-slot catch-up obligation in 2024 because much of the observed fleet is already near or above that cap. It is a stress case, not a recommended lifetime.

The 17-year scenario advances 21,908 slot-years relative to the assumed chassis-life deadlines, compared with 334 under the 14-year scenario. These are remaining-service exposures, not monetary losses: resale and ordinary-renewal cash flows are needed to value them. The benchmark provides a numerical ten-year procurement-capacity test while retaining the operational feasibility gate.

Year	Converted slots	EV renewals	Non-electric slots left
2024	775	0	6,918
2025	776	0	6,142
2026	776	0	5,366
2027	776	0	4,590
2028	776	0	3,814
2029	776	0	3,038
2030	775	1	2,263
2031	770	6	1,493
2032	769	7	724
2033	724	52	0

Table 9: An optimal London 17-year-cap fleet-slot roadmap. Like-body available slots are maintained; this table does not assign routes or chargers.

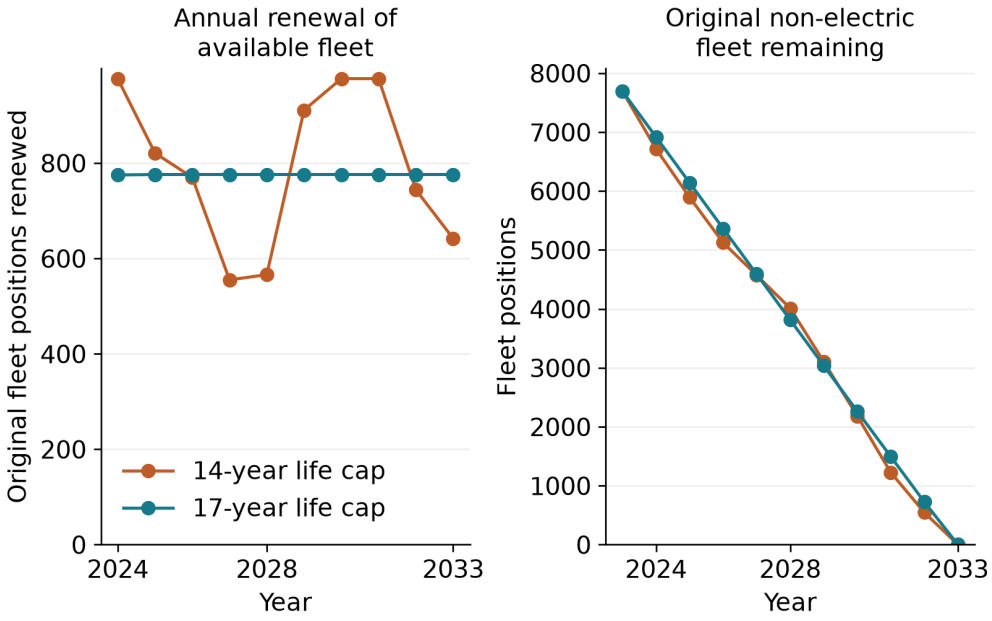


Figure 4: Executed London age-cohort roadmaps under fixed availability. The displayed numbers require compatible electric capacity, all-electric accessories, adequate chargers and trip checks before they become actual bus-purchase quantities.

7 Sensitivity, limitations and decision boundary

Reliable range is the dominant operational variable in the executed screen. At 149 km for the 40-ft proxy and 150 km for articulated buses, the weekday timetable requires 2,444 sequential battery

requests versus 1,502 participating blocks, a 62.7% increase. At 250/251 km it requires 1,901 (26.6%), and at 300 km it requires 1,741 (15.9%). Even the 350-km sensitivity still creates 1,623 requests because several long duties exceed that distance. A 745-km highway block is the longest observed duty and necessarily needs terminal charging, a vehicle exchange or block redesign. These are screening bounds, not physical fleet counts.

To test whether the named cost score is a fragile point choice, we vary seven documented inputs jointly: the 1,711–1,733 fleet bridge, seven paired range cases, 5–18 assumed existing BEBs, 9/17–11/17 baseline renewal, a 1.5–5.0% discount rate, the hardware versus project-envelope vehicle basis, and four baseline fuel intensities. The resulting 432,768-case Cartesian grid reproduces the named CAD 4.575-billion, −96.4% cell exactly. Across the equal-weight assumption grid, median gross PV is CAD 5.26 billion and median economic and environment scores are −127.2% and 33.4%; the joint 5th–95th spans are −166.9 to −89.8% and −15.1 to 41.5%. Range/request count has the largest rank association with economic score, followed by the project-envelope cost basis. These are scenario quantiles, not sampling confidence intervals or probabilities of real outcomes.

Funding does not alter resource cost. In the generic case, 0/25/50% eligible grants change the agency present value from CAD 4.58 to 3.64 and 2.71 billion, and the score from −96.4% to −56.4% and −16.4%. The class-specific central case gives −75.0/−42.8/−10.5%; even its 50% grant leaves agency PV CAD 0.283 billion above the CAD 2.693-billion renewal comparator. These grant cases cover incremental conversion capital only; the alternative common-trolley grant calculation holds the comparator unsubsidized and cannot establish parity under symmetric eligibility. The transition therefore needs a stated environmental/policy value or lower bids, not a claim of financial break-even. The conventional O&M mid-rate is 2.77 CAD/vehicle-km; moving that rate to the hybrid 1.97 or diesel 3.30 value changes the class score to −115.1% or −57.6%. As a separate regional layer, 2023 Bus Operations implies a CAD 0.62–0.97 billion gross-value-added proxy, CAD 0.62–0.79 billion labour income and 6.0–8.8 thousand FTE; these cover the published Bus Operations boundary, not fixed-route service alone. The provided infrastructure cases imply CAD 0.62–3.86 billion of gross construction-period value added over 2024–31, divided across eight years. It is a transition memo, not an additive annual benefit, and construction spending is not credited again as output.[13, 40, 41]

The environmental result is sensitive to the baseline fuel mix, but its sign is not robust: the generic ramp scores 0.7%, 33.7%, 38.0% and 41.5% for the four fuel cases. Replacing just one 162-bus cohort of 335-kWh batteries adds 3.148 kt, exceeding the low-carbon case's 2.743-kt saving and making that score negative. This is a stress case, not a predicted replacement. The inventory excludes charging/depot construction and end-of-life credits. Its 0.302–1.58 kg CO_2e/km baseline range and 19 g/km electric operating rate are sourced, while the vehicle and battery factors are transferred from a Latin American example. Adding infrastructure and a battery replacement to a fully matched boundary would lower the score.

The remaining uncertainties are specific and material: a complete January–August 2023 timetable or as-operated records; block-to-vehicle and block-to-garage links; vehicle ages and payload; depot parking, charger ports, firm power and commissioning dates; terminal exchange times; trolley feeder capacity; and 2033 vehicle winter tests. Without them, no city-wide Pareto point can be certified. The class cost also omits battery replacements, retained pre-2024 asset terminal values, resale receipts and infrastructure terminal value; these must be added before calling it a complete total cost of ownership. The current result is a conditional inventory and cost case with an executed feasibility screen, not a claim that every 2033 trip is schedulable.

The three numerical implementation gates are: every scheduled trip is retained; every candidate block satisfies class, capacity, transfer, winter SOC and charger/site checks; and every cost and emission term has a source, year, unit and uncertainty range. Code currently passes the first structurally and returns `UNVERIFIED` when the second or third lacks inputs.

7.1 Conclusion

The model supports an accelerated class-specific pathway but not an unconditional 2033 procurement order. An assumed 1,647-route-BEB conversion plus 262 common trolley renewals zeros conventional stock arithmetically; service preservation is unproved, and 18 initial BEBs are assumed. Its central price-mixed economics are CAD 4.71 billion versus CAD 2.69 billion for ordinary renewal plus operation, and its partial-production environmental reduction is 4.6–42.7%. Double-deck and shuttle duties require certified terminal or en-route charging before those purchase counts become service-feasible. The defensible near-term decision is to begin grid and depot work immediately, procure proven heavy buses first, and make each later cohort contingent on published charger, SOC, capacity, winter-test and cost certificates.

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Letter to a Metro Vancouver decision-maker

To: TransLink Board and Mayors' Council
Subject: Conditions for a service-preserving all-electric fixed-route bus fleet by 2033

Dear decision-makers,

We treat 2033 as an accelerated target against TransLink's published 2040 goal. The recovered fall-2023 GTFS schedules 54,497 fixed-route bus trips across 25 service patterns; the representative weekday has 1,752 source-declared duty blocks and 1,087 simultaneous trips. We would retain every departure, route direction and capacity proxy in that timetable. Existing trolleybuses remain part of the electric baseline.

Under a 300-km overnight-charge sensitivity, the weekday timetable creates 1,741 sequential battery-block requests for 1,502 participating blocks, 15.9% more duty requests. A 700-kWh double-decker still leaves 17 of 57 duties short of one overnight charge, and a 150-kWh shuttle leaves 49 of 101. These are screening positions, not purchased bus counts: garage links, terminal exchange times, payload and site power are not public.

In the conditional inventory scenario, 1,647 route BEBs replace 1,445 conventional buses and add 202 duty positions; 262 further electric purchases renew the already-electric trolley fleet one-for-one. Gross present value is CAD 4.71 billion against CAD 2.69 billion, an economic score of -75.0%. A 25% grant gives -42.8%; a 50% grant gives -10.5%. The gross resource cost is unchanged, and the high class envelope reaches CAD 5.26 billion (-95.5%). Its 1,927-electric-asset endpoint is stock arithmetic, not a service-feasibility certificate.

The class-specific partial-production, operating and maintenance boundary yields a 4.6–42.7% emissions reduction. The 2023 fixed-route tailpipe proxy is 126–408 tonnes NO_x, 1.11–8.34 tonnes $PM_{2.5}$ and 2.34–14.18 tonnes PM_{10} per year. Depot construction and end-of-life credits remain excluded. The published Bus Operations umbrella is also associated with CAD 0.62–0.97 billion of annual regional value-added and 6.0–8.8 thousand FTE; construction-period effects are separate and non-additive.

Start depot design, utility interconnection and permits in 2024; the published delivery range is six to eight years. Determine purchase quantities through dispatch and commissioned-site checks, test 40- and 60-foot buses under fully electric winter conditions, and hold double-deck and shuttle conversion behind in-route charging and verified payload. If a block fails time, SOC, charger or site capacity, publish that exact gap rather than deleting service.

Sincerely,
Fleet Transition Study Team

Report on Use of AI

Tools and versions. OpenAI Codex (GPT-6) and deepseek-flash were used in later sessions for research planning, source discovery, code implementation, data audits, mathematical review, and English drafting. No original 2023 HiMCM solution paper was requested or used.[50]

Queries and work products. The prompts requested: (i) an evidence-locked plan for environmental, financial and three-city fleet-transition models; (ii) independent literature and primary-source discovery, explicitly excluding the 2023 HiMCM solution; (iii) implementation and testing of trip/block, reserve, charger and tariff models; (iv) recovery and audit of public 2023 TransLink GTFS; (v) a high-standard LaTeX manuscript and reproducible repository; and (vi) completion of missing inputs without invented provenance. The corresponding outputs are retained in `src/`, `data/processed/`, `results/`, this manuscript, and the GitHub history. `research/ai_query_log.md` retains the original-language request texts and their artifact mapping; `research/ai_use_report.md` records the broader source and verification boundary.

Verification. Source URLs, dates, pages, units and hashes are retained in the parameter ledger and provenance files. GTFS files match immutable Git blobs. The model suite includes explicit tests for schedule time, 2033 electrification, energy reserve, charger concurrency, tariff bounds, cost transfer and cohort feasibility. Computed values in this paper are traceable to named scripts. The human team must still verify every cited fact and decide what to submit.

Request and output index

The original Chinese request texts are retained without paraphrasing in `research/ai_query_log.md`. The following is the working English summary of each request and its complete retained artifact set; internal reasoning traces are not treated as research outputs.

1. *Plan the study, choose a well-documented metropolitan case with fallback cities, and search independent electric-bus/fleet/infrastructure literature without using the 2023 HiMCM solution.* Outputs: research boundary, literature notes, parameter ledger, and the three-city design.

2. *Use LaTeX, research other-year award-winning paper presentation, standardize the format, and run the calculations on the available CPU.* Outputs: the canonical TeX/PDF, GTFS and schedule audits, charging LP, London MILP, 432,768-case grid, scenario results, figures, and format checks.

3. *Fill missing fleet, infrastructure, double-deck/shuttle, regional economic and air-quality evidence while refusing to invent unavailable records.* Outputs: APTA negative-evidence audit, deck/shuttle technology audit, regional/NOx/PM audit, class-specific cost bound, updated ledger and manuscript.

4. *Verify all figures and tests, preserve reproducibility, and synchronize the project to the authorized private GitHub repository when access allows.* Outputs: the local Git history, `scripts/reproduce_local.py`, manuscript-number and PDF checks, and the repository-ready file set.